

Development Of Synthetic Clots With Shear-Induced Fragmentation For Improved In Vitro Neurointerventional Device Testing

Wylie Ummel¹, Ashley Wintermantel¹, Nathan Hartman¹, Kailey Lewis^{1,2}, Jesse Wells^{1,2}, Steven Schwartz II¹, Sophia Robertson, PhD.^{1,2}, Wyatt Clark, PhD.^{1,2}, Tim Becker, PhD.^{1,2}

Northern Arizona University – Mechanical Engineering Department¹, Aneuvast Technologies Inc.²

Abstract

Ischemic stroke is a leading cause of death and long-term disability, with mechanical thrombectomy now an established standard of care. A need exists for validated synthetic clot analogs for in vitro thrombectomy testing and device comparison. While existing synthetic clot analogs accurately reproduce the compressive and shear mechanical properties of human thrombus, they generally fail to replicate clinically relevant clot fragmentation and distal embolization observed during thrombectomy. **We present a synthetic thrombus with tunable, physiologically accurate fragmentation behavior and quantified particulation.** Under aspiration (~77 kPa) in a Circle of Willis flow model (CW), these analogs particulate and embolize distally after an initial pass, often requiring a second aspiration. No known existing synthetic clot analog reproduces this sequence of clinically relevant failure behaviors. Clot failure: surface erosion, creep rupture, shear and tensile fracture, and ingestion or corking with or without particulation. The thrombi remain visible under ultraviolet light, enabling direct visualization of clot-device interaction and quantitative in vitro assessment of first-pass effect (FPE).

Introduction

Mechanical thrombectomy preclinical device evaluation remains limited by the fidelity of synthetic clot analogs. Existing analogs are validated under published compressive and shear moduli, metrics that describe resistance to deformation but not fragmentation behavior.^{1,2,3} Clinically significant shear-induced clot fragmentation can drive distal embolization and influence first-pass effect (FPE), particularly in platelet-rich thrombi. Previous work from the Bioengineering Devices Lab (BDL) developed a synthetic thrombus with reproducible, physiologically relevant mechanical properties but limited shear-induced fragmentation. **This study introduces controlled internal heterogeneities into the established synthetic clot that produce tunable and reproducible fragmentation while preserving physiologic mechanical behavior.**

Methods

Clot fabrication: Thrombi are fabricated from a polyacrylamide–alginate (PAAM–Alg) composite viscoelastic hydrogel. Previously developed hard clot formulation is granulated and encased in a soft clot matrix. Adjusting the size, shape, clot formulation, and ratio of the composite produces a repeatable and tunable particulating clot. (Figure 1).

Mechanical characterization:

Shear and compressive moduli are measured with testing plates on a DMA-Rheometer (TA Instruments). Shear failure stress, the stress at which clots shear apart, is determined by a controlled plate rotation (10°/sec) to record clot shear separation (Figure 2).

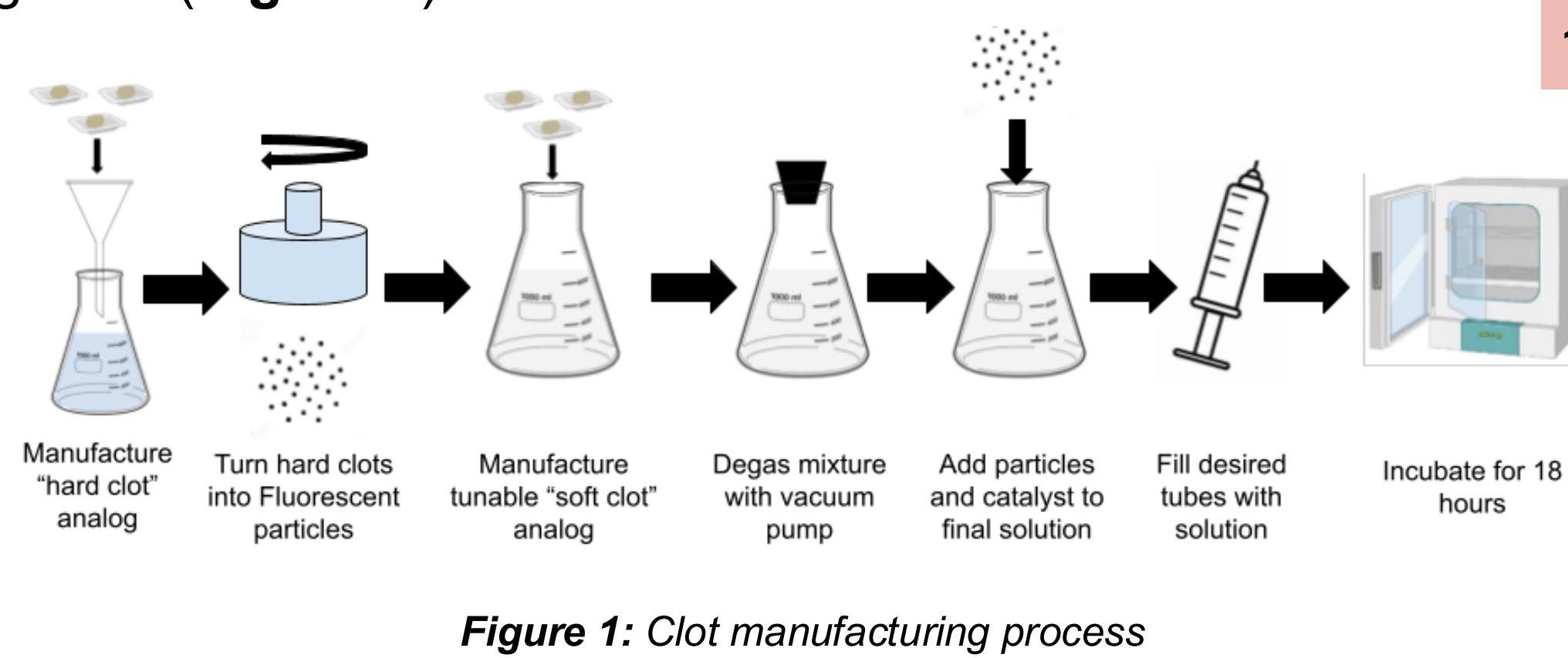


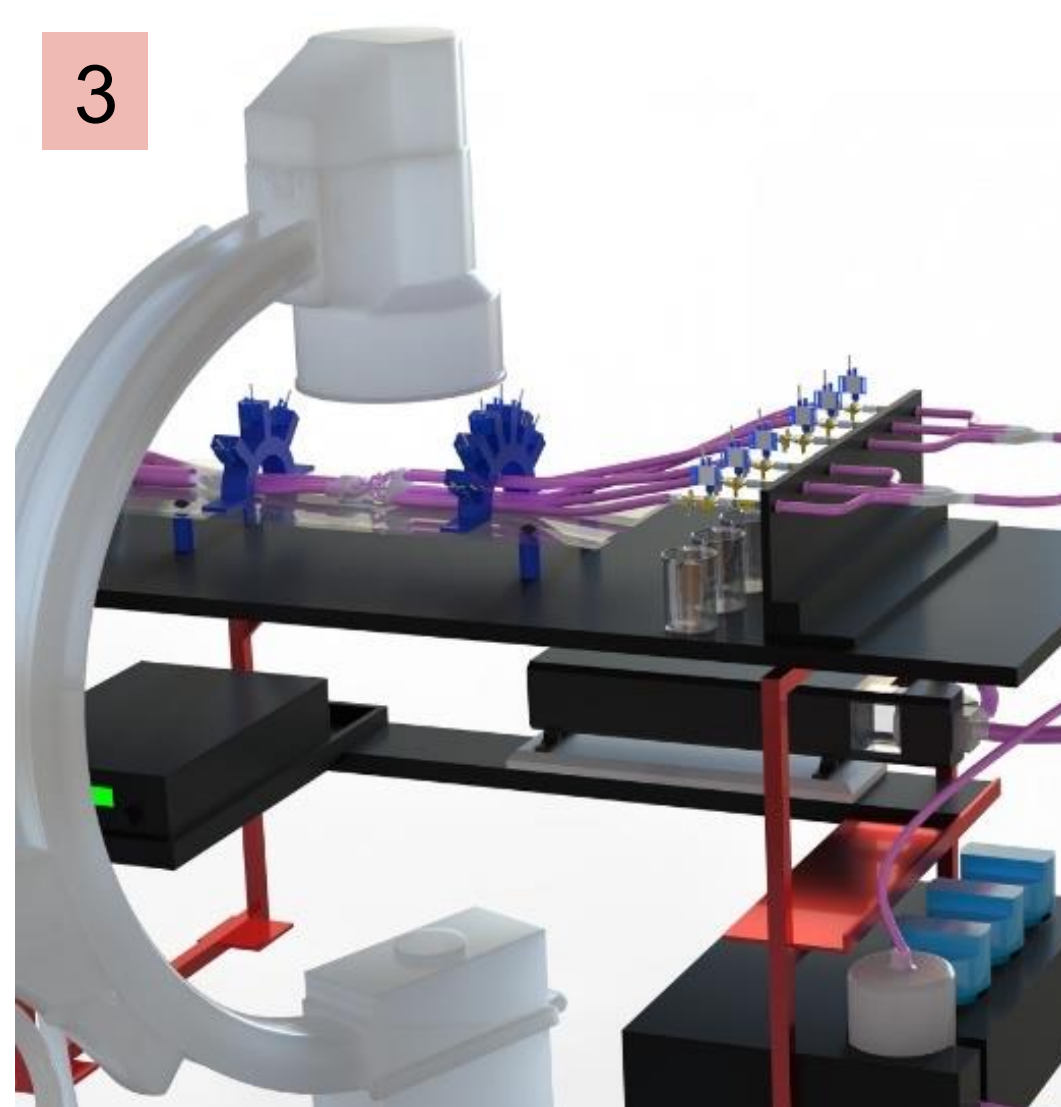
Figure 1: Clot manufacturing process

- Clots were compressed to 50% before running the test
- Rheometer spins at 10°/sec for 540°, typically fracturing multiple times throughout a full rotation
- All mechanical data was compared to published human clot data^{2,3} (Figures 4 and 5)

Figure 2: Clot analog loaded on parallel-plate rheometer for shear testing.

Aspiration testing: Utilizing the simulated surgical suite (BDL-S³), particulating clots are introduced into the ICA and aspirated with a Zoom 71 (Imperative Care) catheter. Based on the observed failure mode, each trial is classified as ingested with particulation, corked without particulation, fragmented with distal embolization, or unrecoverable distal fragmentation (Figure 3).

Figure 3: Bioengineering Devices Lab's Simulated Surgical Suite (BDL-S³, Northern Arizona University) – includes a clinically relevant in-vitro model, a programmable pulsatile pump (Super Pump AR-Series, ViVtiro Labs), shear-thinning viscosity-matched blood analog, physiologically matched 3D-printed circle of Willis (CW) model, CW branch pressure recording (LabVIEW), hemodynamic flow visualization, and C-arm fluoroscopic imaging.



Results

Quantification of clot properties:

Particulating and Soft Clot Analogs Exhibit Similar Moduli Within the Physiological Range:

- **Compression:** 14.7 kPa (control), 11.4 kPa (particulating) (Figure 4)
- **Shear:** 3.6 kPa (control), 2.8 kPa (particulating)
- Black boxes indicate the clinically relevant range of previously published clot properties

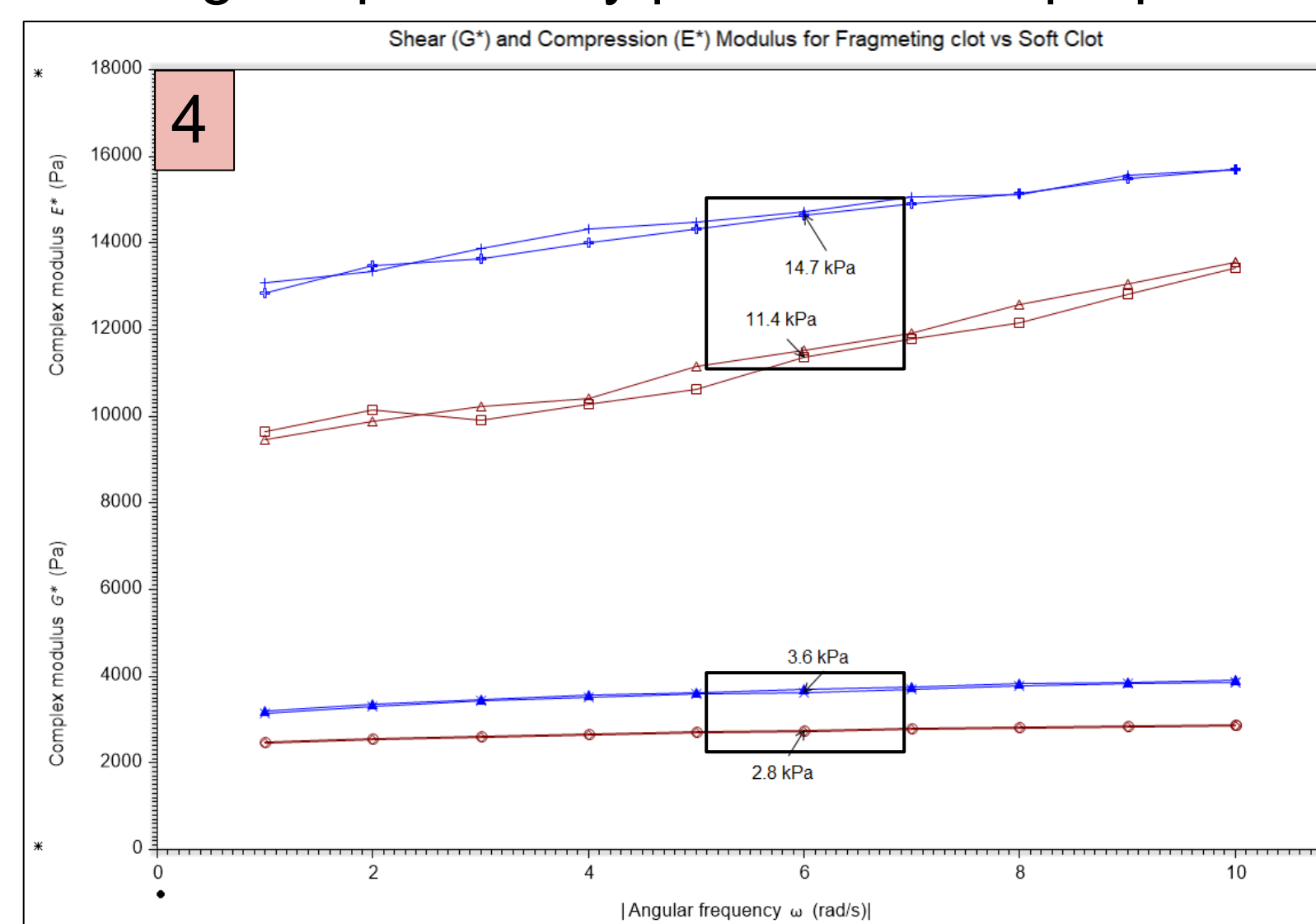


Figure 4: Shear and compression results of both the soft clot control and particulating clot

Particulating Clot Analog Reproduces Physiological Fragmentation Behavior Compared with a Non-Fragmenting Control:

- Soft clot control reaches a peak stress of 12.1 kPa then returns to zero as it dislodging from fixture (Figure 5)
- Particulating clot analog reaches a lower peak stress (7.0 kPa), holding 40-70% of peak through entire rotation period.

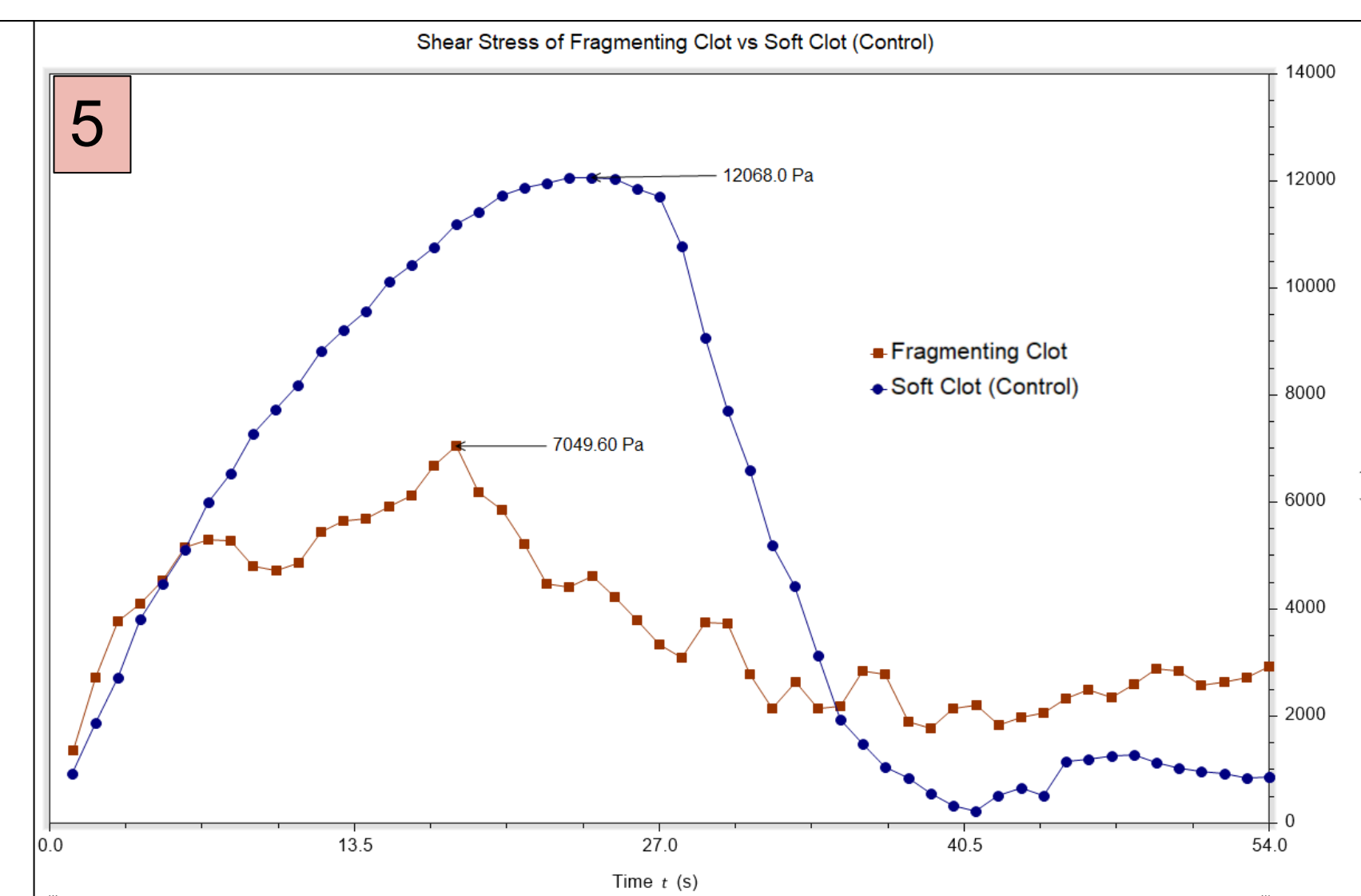


Figure 5: Shear failure results with peak stresses of both the soft clot control and particulating clot

Aspiration of particulating clot, 1st attempt - resulting in distal embolization:

Thrombi lodged in the internal carotid artery (ICA) under physiological flow (Figure 6B) failed to completely ingest during first aspiration attempt, with ~50% of vacuum pressure lost at the catheter tip. The clot fragmented and embolized the distal M1-M2 bifurcation (Figure 7A). This secondary embolization event is recorded in real-time as fluctuations in CW branch pressures plot (Figure 6C). The upper panel illustrates the real-time drop in pressure at the M2 branch as the clot becomes lodged (purple waveform: 6C – left side)

Homogeneous controls produced no distal emboli. **The particulating analog is the only substrate in which first-pass success and distal embolization both occur**, making FPE measurable.

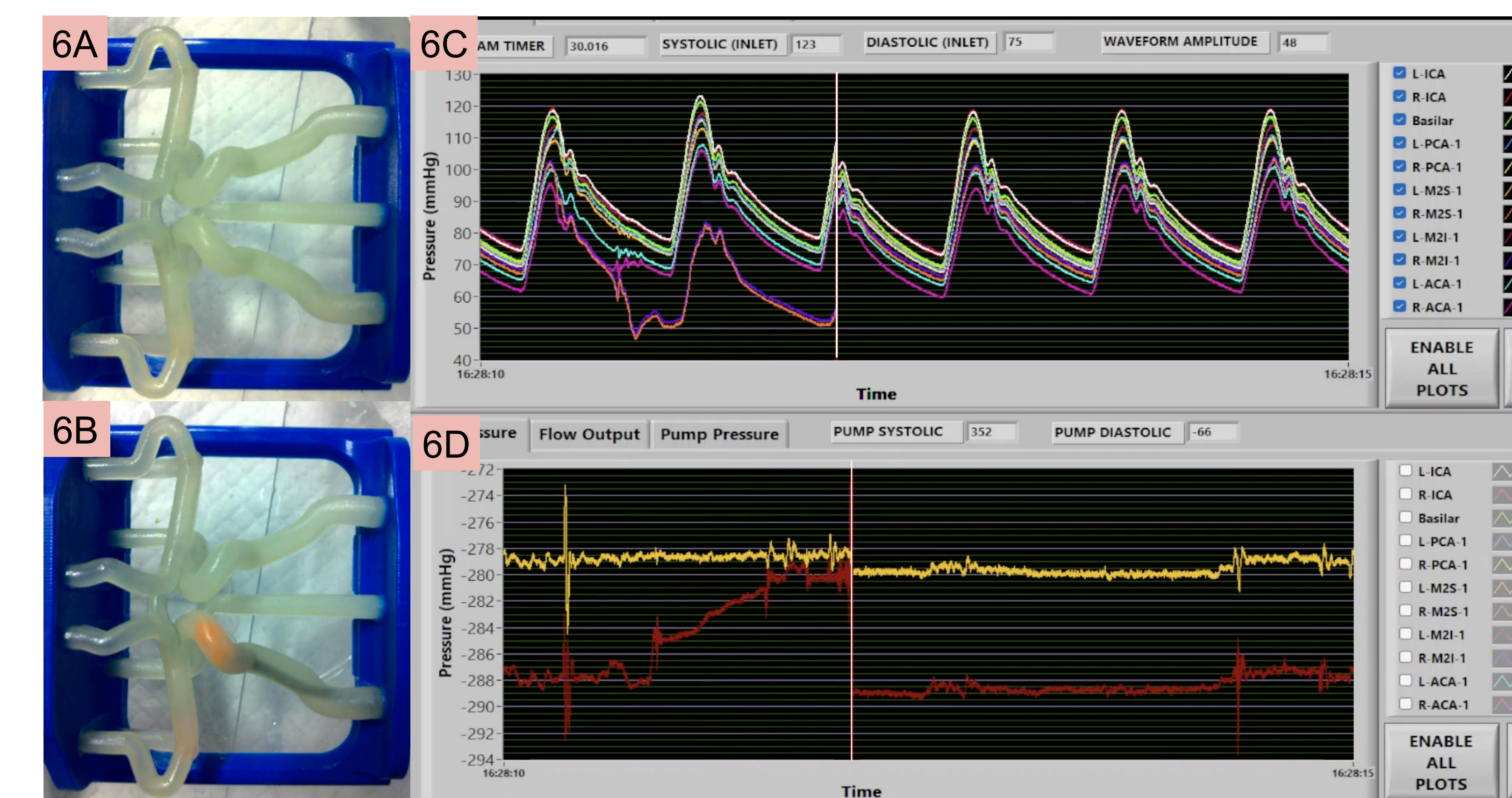


Figure 6: A) Cleared Circle of Willis (CW) model prior to clot introduction. B) Synthetic thrombus lodged in the right ICA C) CW pressures across all branches (L/R-ICA, basilar, L/R-PCA, L/R-M2S, L/R-M2I, L/R-ACA) under pulsatile pressures. D) Catheter (red) and syringe (yellow) real-time pressure readings during aspiration vacuum (50% loss, normal vacuum ~ -600 mmHg).

The fluorescent tracer (PIV particles) shows where clot material migrates: retrieved into the syringe (8A), Distal fragments recovered downstream (8B), or particulation with distal embolism (captured within CW 7A).

Results - Continued

Aspiration of particulating clot, 2nd attempt - resulting in complete clot removal:

Clot fragment engaged at M2 bifurcation. Distal aspiration was attempted (Figure 7A). Decreased pressure verifies low/no flow in M2 (purple waveform: 7C – left side). Aspiration initiated: real-time pressure drop at the catheter hub (red), matching syringe vacuum pressure (yellow) (7D), at the same time point: clot is aspirated and pressure restored in (7C – right side). M2 and collateral flow restored to nominal (7B).

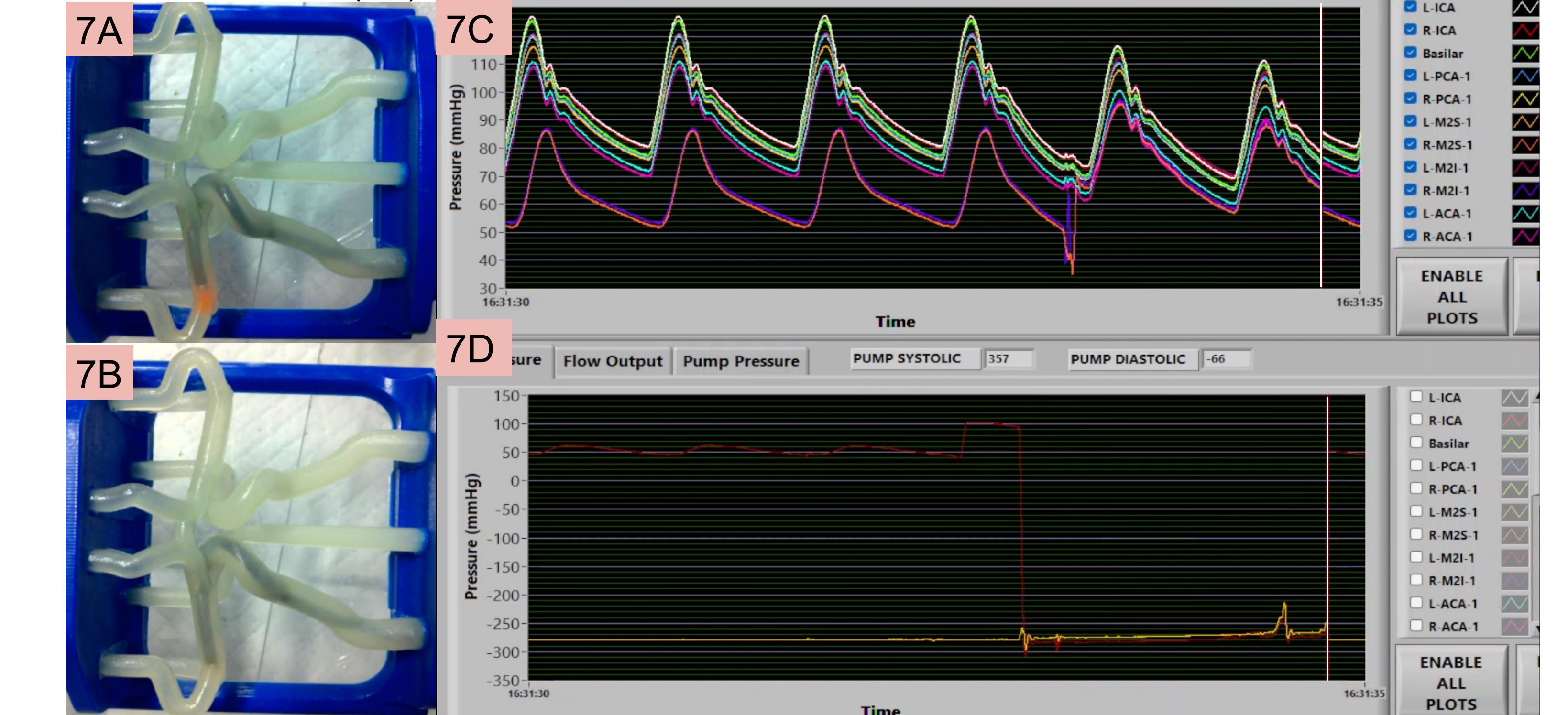


Figure 7: A) Clot engagement at distal embolization. B) Full aspiration of clot C) real-time pressure data of all branches. D) pressure at the syringe (yellow) and catheter hub (red) throughout aspiration.

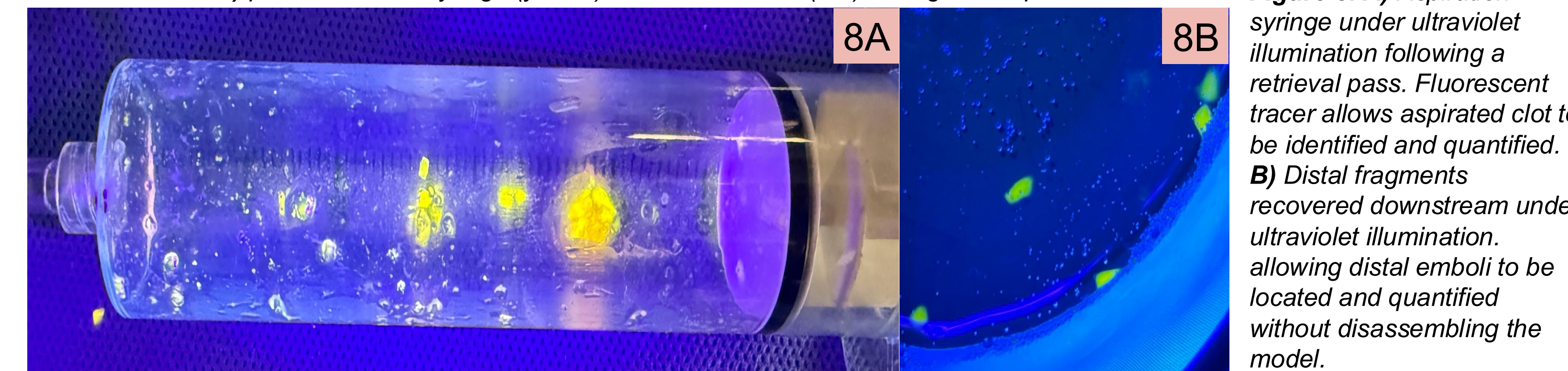


Figure 8: A) Aspiration syringe under ultraviolet illumination following a retrieval pass. Fluorescent tracer allows aspirated clot to be identified and quantified. B) Distal fragments recovered downstream under ultraviolet illumination, allowing distal emboli to be located and quantified without disassembling the model.

Discussion & Conclusion

Unlike conventional soft-clot polymer analogs that resist particulation, the proposed analogs fragment under applied load due to their composite structure—from a discontinuous hard-particle phase dispersed within a soft matrix. Demonstrating fragmentation is an independent mechanical property that cannot be predicted by modulus alone. A shear test was created to verify the clinically relevant fracture modes. During flow model testing, clot analogs were ingested, corked at the catheter tip, or fragmented with distal migration, including reproducible M1-M2 bifurcation emboli. **Capturing this spectrum of retrieval outcomes enables meaningful in vitro measurement of the first-pass effect and supports specifying clot analogs by both modulus and fragmentation behavior.** Current thrombectomy devices are validated using synthetic thrombi that cannot fail the way human thrombi fail. Limitations include testing at room temperature and the absence of direct comparison between fragment distributions and retrieved human thrombi.

Future Studies

- Compare fragment-size distributions to platelet-rich blood clots under matched aspiration settings.
 - Establish the analog as a standardized test article for verifying and validating thrombectomy and embolic-protection devices
 - Incorporate radiopacity (barium sulfate) to expand imaging capabilities
- Scan the QR code on the right to visit the Bioengineering Devices Laboratory (BDL) website. Website includes research, publications, lab resources, and contract testing/manufacturing capabilities.

References

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